

UNIVERSITY OF TRANSPORT HO CHI MINH CITY

INFORMATION SECURITY

CLASSICAL CIPHERS

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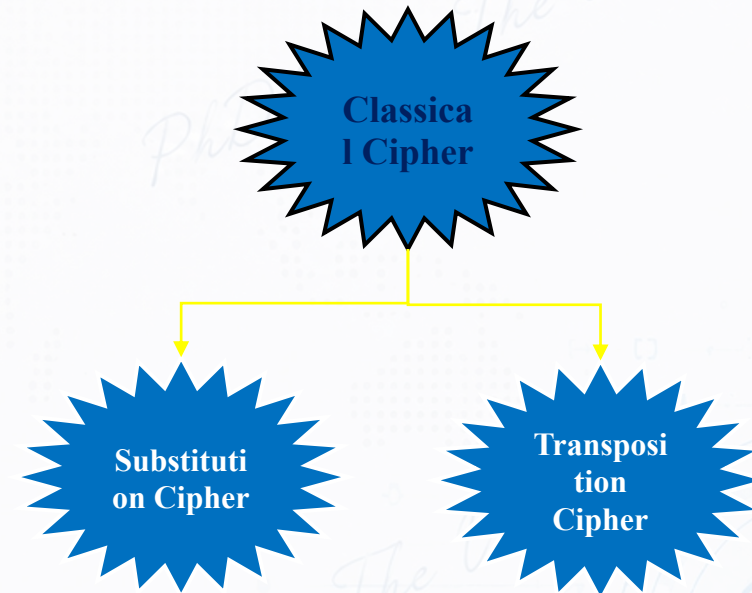
CLASSICAL CIPHER

Classical Cryptosystems

- All classical cryptosystems are based on the **symmetric-key encryption paradigm**.
- The primary security service provided by these systems is **confidentiality**.
- Unlike modern cryptographic systems, which operate on **binary data**, classical cryptosystems operate on an **alphabet** as the fundamental unit of processing.

Two main categories of classical ciphers:

1. **Substitution Cipher**
2. **Transposition Cipher**



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Substitution Cipher

In a substitution cipher, each character of the plaintext drawn from a fixed alphabet is replaced by another character from the same alphabet according to a substitution rule determined by the key. For example, in a Caesar cipher with a shift of 2, the letter A is replaced by C, B is replaced by D, and so on.

Substitution ciphers are classified into two main categories:

- **Monoalphabetic Ciphers**
- **Polyalphabetic Ciphers**

↑ A B C D E F G H I J K L M
X Y Z A B C D E F G H I J

↑ N O P Q R S T U V W X Y Z
K L M N O P Q R S T U V W

ciphertext:
ZLLI EXQ.
IRKZE PLLK?

plaintext:
COOL HAT.
LUNCH SOON?

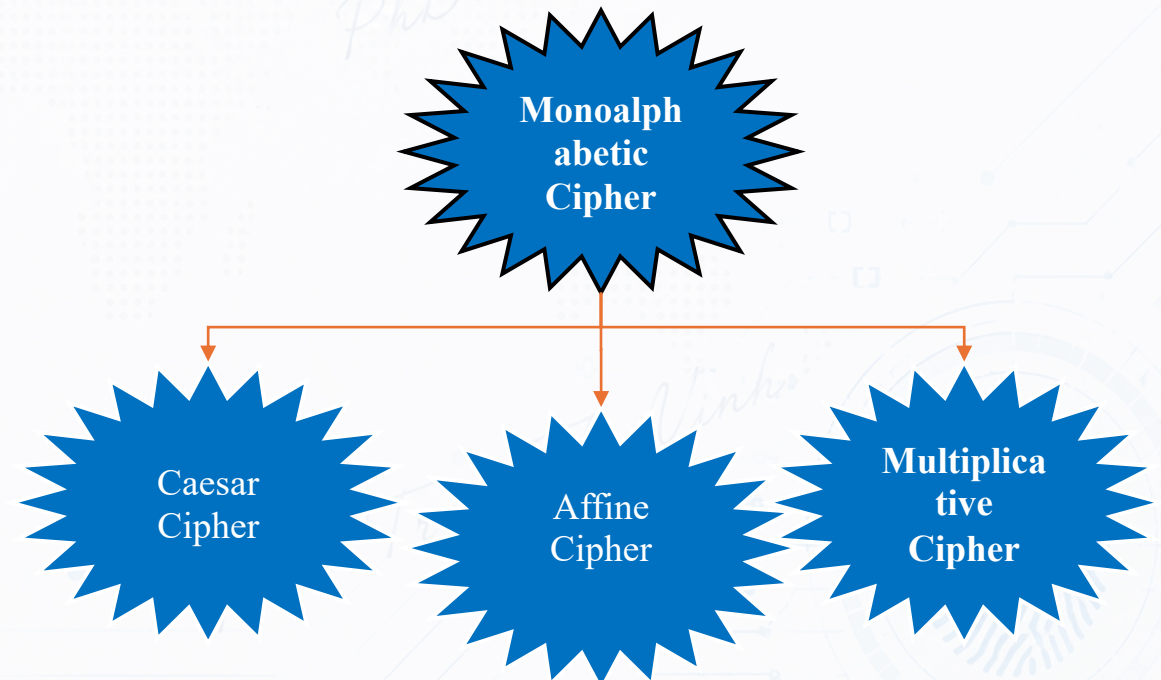
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Monoalphabetic Cipher

In a monoalphabetic cipher, each plaintext character is mapped to a unique ciphertext character. Consequently, regardless of how many times a particular character appears in the plaintext, it is always replaced by the same ciphertext character. As a result, the plaintext and ciphertext have the same length.

Types of monoalphabetic ciphers:

- **Shift Cipher**, also known as the **Caesar Cipher** or **Additive Cipher**
- **Multiplicative Cipher**
- **Affine Cipher**



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Polyalphabetic Cipher

In a polyalphabetic cipher, a plaintext character is not necessarily mapped to the same ciphertext character each time it appears. Instead, the substitution depends on the character's position or the state of the key during encryption. Consequently, multiple occurrences of the same plaintext character may be encrypted into different ciphertext characters.

This property reduces the effectiveness of **frequency analysis**, a common cryptanalytic technique used against monoalphabetic substitution ciphers.

Types of polyalphabetic ciphers:

- **Autokey Cipher**
- **Vigenère Cipher**
- **Playfair Cipher**
- **Hill Cipher**
- **One-Time Pad (OTP)**
- **Rotor Cipher**

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Transposition Cipher

Unlike a substitution cipher, a transposition cipher does not replace plaintext characters with different characters. Instead, it preserves the original characters and rearranges their positions according to a predetermined rule. Consequently, a character appearing at one position in the plaintext may appear at a different position in the ciphertext.

Types of transposition ciphers:

- **Columnar Transposition Cipher**
- **Rail Fence Cipher**

SUBSTITUTION CIPHER - CAESAR CIPHER

The Caesar cipher is one of the earliest recorded classical ciphers. It is traditionally attributed to Julius Caesar, who reportedly used it for military correspondence.

In a Caesar cipher, each plaintext letter is replaced by another letter located k positions further along the alphabet. The shift operation is performed modulo the size of the alphabet.

The encryption function is defined as:

$$E_k(j) = (j + k) \bmod n$$

where:

- j is the index of a plaintext character in the alphabet.
- k is the shift key.
- n is the size of the alphabet.
- $E_k(j)$ is the index of the corresponding ciphertext character.

For example, using the English alphabet ($n = 26$) and a shift key of $k = 3$:

$$A \rightarrow D, B \rightarrow E, \dots, X \rightarrow A, Y \rightarrow B, Z \rightarrow C$$

SUBSTITUTION CIPHER - CAESAR CIPHER

Therefore, the key in the Caesar cipher is the value k , which determines the magnitude of the shift. The decryption function is the inverse of the encryption function:

$$E_k^{-1}(j) = E_{n-k}(j) = (j + n - k) \bmod n$$

or equivalently,

$$D_k(j) = (j - k) \bmod n$$

where $D_k(j)$ denotes the decryption function.

If necessary, the alphabet can be extended to include uppercase letters, lowercase letters, digits, punctuation marks, and other special characters, allowing the cipher to process the entire character set of the source text.

For an alphabet of size n , there are n possible shift keys. One of these keys, namely $k = 0$, corresponds to the **identity transformation**, meaning that the ciphertext is identical to the plaintext.

SUBSTITUTION CIPHER – KEYWORD CAESAR CIPHER

Keyword Caesar Cipher

In this variant of the Caesar cipher, the key consists of a shift value k ($0 \leq k \leq n - 1$) and a keyword or short phrase.

To construct the substitution alphabet, the plaintext alphabet is first written out. Starting at position k , the letters of the keyword are inserted, with duplicate letters removed. The remaining positions are then filled with the unused letters of the alphabet in their natural order.

The resulting alphabet defines the substitution table used for encryption.

Keyword: “yes” , $k = 2$	
Plain Alphabet	a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z
Cipher Alphabet	x, z, y, e, s, a, b, c, d, f, g, h, i, j, k, l, m, n, o, p, q, r, t, u, v, w
Plaintext	data
Ciphertext	expx

For decryption, the receiver must know both the shift value k and the keyword in order to reconstruct the substitution alphabet. The reverse substitution is then applied to recover the original plaintext.

SUBSTITUTION CIPHER – AFFINE CIPHER

The Affine cipher is a **monoalphabetic substitution cipher** in which each plaintext character is transformed into a ciphertext character using a linear function over modular arithmetic.

The characters of an alphabet of size m are first mapped to the integers 0 through $m - 1$.

The key consists of two integers, a and b , subject to the condition:

$$\gcd(a, m) = 1$$

This condition ensures that a has a multiplicative inverse modulo m , which is required for decryption.

Encryption Function

$$E(x) = (ax + b) \bmod m$$

where:

- x is the numerical representation of a plaintext character.
- m is the size of the alphabet.
- a and b are the key parameters.
- $\gcd(a, m) = 1$.

Decryption Function

To recover the plaintext, the inverse transformation is applied:

$$D(x) = a^{-1}(x - b) \bmod m$$

where

$$a^{-1}a \equiv 1 \pmod{m}$$

and a^{-1} denotes the multiplicative inverse of a modulo m .

SUBSTITUTION CIPHER – AFFINE CIPHER

Frequency Analysis of Monoalphabetic Ciphers

The Caesar cipher and the Affine cipher both belong to the class of **monoalphabetic substitution ciphers**. This means that, for a fixed key, each plaintext character is always replaced by the same ciphertext character.

As a result, these ciphers are vulnerable to **frequency analysis**. This cryptanalytic technique exploits the fact that substitution preserves the frequency distribution of characters while merely changing their labels.

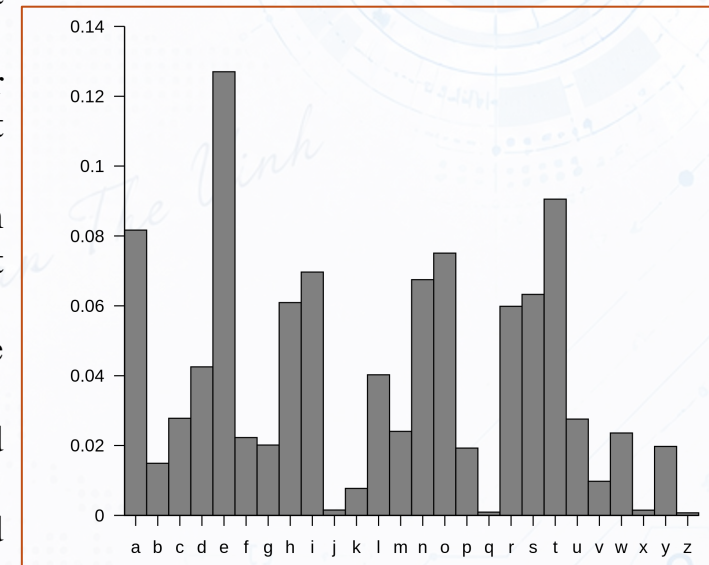
In natural languages, character frequencies are not uniformly distributed. For example, in English text, the letter **E** occurs with a frequency of approximately 12.7%, whereas the letter **V** appears with a frequency of only about 1%.

By computing the frequency distribution of characters in the ciphertext and comparing it with the known frequency distribution of the corresponding language, an attacker can infer probable mappings between ciphertext and plaintext characters.

For the Caesar cipher, identifying one or a few corresponding character pairs is often sufficient to recover the shift key k .

For the Affine cipher, at least two character correspondences are required to establish a system of equations and determine the key parameters a and b .

For the **Keyword Caesar Cipher**, the effectiveness of frequency analysis depends largely on the length and structure of the keyword. However, since it remains a monoalphabetic substitution cipher, it is still vulnerable to frequency-analysis attacks when a sufficient amount of ciphertext is available.



This weakness led to the development of polyalphabetic ciphers such as the Vigenère Cipher, where the same plaintext character may be mapped to different ciphertext characters, making frequency analysis significantly more difficult.

SUBSTITUTION CIPHER – VIGENÈRE CIPHER

In the sixteenth century, the French diplomat **Blaise de Vigenère** popularized a polyalphabetic substitution cipher that later became known as the **Vigenère Cipher**.

In this cipher, the key consists of a word or phrase containing d characters. During encryption, the key is repeated cyclically until it matches the length of the plaintext.

Each plaintext character is encrypted by combining it with the corresponding key character using modular addition over the alphabet.

The encryption function is defined as:

$$c_i = (m_i + k_i) \bmod n$$

where:

- m_i is the index of the i -th plaintext character.
- k_i is the index of the corresponding key character.
- c_i is the index of the i -th ciphertext character.
- n is the size of the alphabet.
- d is the length of the key.

The key is applied cyclically:

$$k_i = k_{i \bmod d}$$

Decryption Function

The plaintext can be recovered by applying the inverse transformation:

$$m_i = (c_i - k_i + n) \bmod n$$

The Vigenère Cipher can be viewed as a generalized Caesar Cipher in which the shift value changes from one character position to another according to a repeating key.

SUBSTITUTION CIPHER – VIGENÈRE CIPHER

Cryptanalysis of the Vigenère Cipher

Plaintext: HELLO

Key: KEYKE

Ciphertext: RIJVS

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

where

$$H + K = R, E + E = I, L + Y = J, L + K = V, O + E = S$$

using modulo 26 addition.

The fundamental difference between the Vigenère cipher and the previously discussed ciphers is that it belongs to the class of **polyalphabetic substitution ciphers**.

In monoalphabetic ciphers such as the Caesar and Affine ciphers, a given plaintext character is always mapped to the same ciphertext character for a fixed key. In contrast, in the Vigenère cipher, the same plaintext character may be encrypted into different ciphertext characters depending on its position and the corresponding key character.

As a result, the frequency distribution of the ciphertext no longer directly reflects the frequency distribution of the underlying language, making simple frequency analysis significantly less effective.

One of the most famous classical attacks against the Vigenère cipher is the **Kasiski Examination**.

In 1863, the German cryptanalyst **Friedrich Kasiski** observed that the key length could be estimated by identifying repeated sequences within the ciphertext.

Suppose two identical ciphertext segments are found and the distance between them is 20 characters. This may indicate that two identical plaintext segments were encrypted using the same portion of the repeating key.

Consequently, the key length is likely to be a divisor of the observed distance.

Since

$$20 = 1 \times 20$$

the possible key lengths are

$$1, 2, 4, 5, 10, 20.$$

Once the key length l has been determined, the ciphertext can be divided into l independent streams.

Each stream corresponds to a Caesar cipher encrypted with a fixed shift value. Frequency analysis can then be applied separately to each stream in order to recover the individual key characters.

SUBSTITUTION CIPHER – VIGENÈRE CIPHER

Composite Vigenère Cipher

To increase the key period and make cryptanalysis more difficult, multiple Vigenère encryptions may be applied using different keys.

Assume that r independent keys are used:

$$K^{(1)}, K^{(2)}, \dots, K^{(r)}$$

The i -th ciphertext character is then computed as:

$$c_i = (m_i + k_i + l_i + \dots + s_i) \bmod n$$

where:

- m_i is the i -th plaintext character.
- $k_i, l_i, \dots, s_i$ are the corresponding characters from different keys.
- n is the size of the alphabet.

Each key has its own period:

$$d_k, d_l, \dots, d_s$$

and is applied cyclically:

$$k_i = k_{i \bmod d_k}$$

$$l_i = l_{i \bmod d_l}$$

$$s_i = s_{i \bmod d_s}$$

For maximum effectiveness, the key lengths should preferably be pairwise coprime.

The resulting overall period of the cipher is:

$$Period = lcm(d_1, d_2, \dots, d_s)$$

where lcm denotes the least common multiple.

If all key lengths are pairwise coprime, then:

$$lcm(d_1, d_2, \dots, d_s) = d_k d_l d_s \dots$$

A significantly longer period makes it substantially more difficult for a cryptanalyst to determine the key length using techniques such as the Kasiski Examination or other statistical attacks.

SUBSTITUTION CIPHER – POLYBIUS SQUARE

The **Polybius Square** is a classical encryption technique belonging to the class of **monoalphabetic substitution ciphers**. It is traditionally attributed to the ancient Greek historian and military strategist **Polybius**.

In this system, the alphabet is arranged in a 5×5 grid. Each character is represented by a pair of digits:

- The first digit denotes the row number.
- The second digit denotes the column number.

Since the grid contains only 25 cells, the letters **I** and **J** are typically combined into a single cell.

Thus, each plaintext character is replaced by its corresponding coordinate pair within the Polybius Square.

Example

Plaintext

TRUONG GIAO THONG VAN TAI TPHCM

Ciphertext

444245343322 22241134 4423343322 511133 441124 4435231332

Decryption is performed by reading each coordinate pair in the ciphertext and mapping it back to the corresponding row and column position in the Polybius Square.

	1	2	3	4	5
1	A	B	C	D	E
2	F	G	H	I,J	K
3	L	M	N	O	P
4	Q	R	S	T	U
5	V	W	X	Y	Z

SUBSTITUTION CIPHER – POLYBIUS SQUARE

Encryption Key

The alphabet is arranged within a rectangular or square matrix. The arrangement of characters in the matrix is kept secret and serves as the encryption key.

To increase the size of the symbol set, digits, punctuation marks, and other special characters may be included. This allows the use of matrices such as:

$5 \times 5, 5 \times 6, 6 \times 6, 5 \times 7, \dots$

depending on the number of symbols to be represented.

Cryptanalysis

The **Polybius Square** remains vulnerable to **frequency analysis**.

Since each plaintext character is consistently mapped to the same coordinate pair, certain statistical characteristics of the underlying language are preserved.

A cryptanalyst may analyze the frequency distribution of coordinate pairs or recurring patterns in the ciphertext and compare them with the known frequency distribution of the target language to infer probable plaintext-to-ciphertext mappings.

Once several characters or character groups have been identified, the remaining message can often be reconstructed through contextual and linguistic analysis.

If the Polybius Square contains n distinct symbols, the number of possible keys (different arrangements of symbols within the matrix) is:

$n!$

For example, a standard 25-character Polybius Square admits:

$25!$

possible keys.

SUBSTITUTION CIPHER – PLAYFAIR CIPHER

Playfair Cipher

The Playfair cipher is a **digraph substitution cipher** in which pairs of characters are encrypted simultaneously rather than individual characters.

The cipher typically employs a 5×5 key square similar to the Polybius Square. The alphabet is arranged within the matrix according to a predefined ordering, with the letters **I** and **J** usually sharing the same cell.

The plaintext is divided into pairs of characters (bigrams), and each pair is encrypted according to one of the following three rules.

Rule 1: Same Column

If both letters of a bigram appear in the same column of the key square, each letter is replaced by the letter immediately below it.

If a letter is located in the bottom row, the letter at the top of the same column is used instead (wrap-around rule).

Example:

$RF \rightarrow FO$

$SY \rightarrow YB$

Rule 2: Same Row

If both letters of a bigram appear in the same row of the key square, each letter is replaced by the letter immediately to its right.

If a letter is located in the rightmost column, the letter in the leftmost column of the same row is used instead.

Example:

$GR \rightarrow RA$

$AM \rightarrow MB$

Rule 3: Rectangle Rule

If the two letters lie in different rows and columns, they define the corners of a rectangle within the key square.

Each letter is replaced by the letter in the same row but in the column occupied by the other letter.

Example:

$EW \rightarrow HU$

B	I	G	R	A	M
C	D	E	F	H	J
K	L	N	O	P	Q
S	T	U	V	W	X
Y	Z		.	,	-

Key: BIGRAM

Plaintext: HELLO-

Ciphertext: JFTTQ.

Key Square Construction

The key square may be generated randomly or derived from a keyword.

When a keyword is used:

1. Write the letters of the keyword into the matrix.
2. Remove duplicate letters.
3. Fill the remaining cells with the unused letters of the alphabet in alphabetical order.

SUBSTITUTION CIPHER – TWO-SQUARE CIPHER

Two-Square Cipher

In 1854, the English scientist **Charles Wheatstone** developed a digraph substitution cipher known as the **Two-Square Cipher**.

The cipher employs two independent 5×5 key squares, each generated from a keyword or a separate alphabet arrangement.

Like the Playfair cipher, the plaintext is divided into pairs of letters (bigrams). However, each letter of the bigram is located in a different key square:

The first letter is found in the left square.

The second letter is found in the right square.

Encryption Rule

Assume that:

The first plaintext letter is located at row r_1 , column c_1 in the left square.

The second plaintext letter is located at row r_2 , column c_2 in the right square.

B	I	G	R	A	M
C	D	E	F	H	J
K	L	N	O	P	Q
S	T	U	V	W	X
Y	Z		.	,	-

W	E	L	C	O	M
A	B	D	F	G	H
I	J	K	N	P	Q
R	S	T	U	V	X
Y	Z		.	,	-

Case 1: Different Rows

The two letters define opposite corners of a rectangle spanning the two squares.

The ciphertext letters are obtained from:

Row r_1 , column c_2 in the left square.

Row r_2 , column c_1 in the right square.

Thus, the ciphertext letters correspond to the other two corners of the rectangle.

Case 2: Same Row

If both letters lie in the same row, each letter is replaced by the letter immediately to its right within its respective square.

If a letter is in the rightmost column, the leftmost character of the same row is used instead.

Key: BIGRAM and WELCOME

Plaintext: **VIETNAM.**

Ciphertext: **RODUIEC-**

SECURITY OF PLAYFAIR CIPHER – TWO-SQUARE CIPHER

The cryptographic strength of bigram ciphers is significantly greater than that of simple monoalphabetic substitution ciphers such as the Caesar and Affine ciphers. Historically, several bigram-based systems, including the Two-Square Cipher, were employed for military communications.

For these ciphers, single-letter frequency analysis is no longer effective. Instead, cryptanalysts must examine the frequency distribution of bigrams or longer character groups.

Although certain bigrams occur more frequently than others in natural languages, the number of possible bigrams is substantially larger than the number of individual letters. Consequently, establishing direct correspondences between plaintext and ciphertext bigrams becomes considerably more difficult.

If the key square contains n distinct symbols, the number of possible arrangements is:

$$n!$$

Thus, the key space of bigram ciphers is significantly larger than that of simple substitution ciphers.

Nevertheless, like other classical cryptosystems, bigram ciphers remain vulnerable to statistical cryptanalysis when sufficient ciphertext is available.

Following the development of substitution ciphers, cryptographic research evolved toward a different class of systems known as **transposition ciphers**, in which the positions of characters are rearranged rather than the characters themselves being substituted.

Substitution ciphers attempt to conceal the identity of characters, whereas transposition ciphers attempt to conceal their positions.

TRANSPOSITION CIPHER – CARDANO GRILLE

Cardano Grille

In the sixteenth century, the Italian mathematician, physician, and philosopher **Gerolamo Cardano** proposed a cryptographic technique based on character transposition, later known as the **Cardano Grille**.

The system employs a grille containing openings cut at predetermined positions. The grille is placed over a square grid used to write the message.

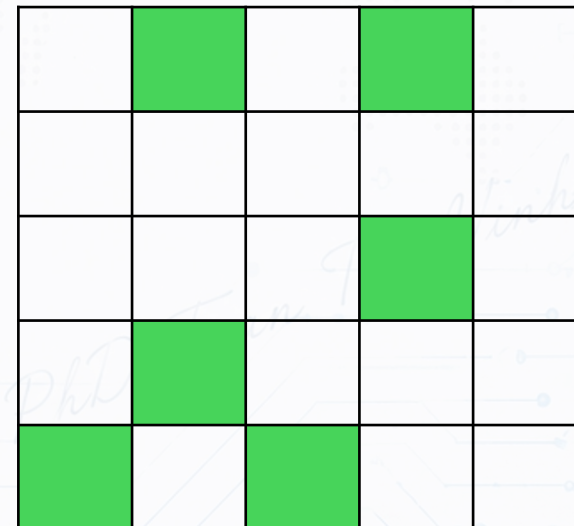
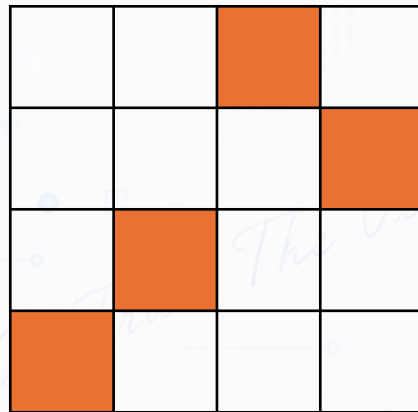
The sender writes plaintext characters into the cells visible through the openings. The grille is then rotated by 90° , 180° , and 270° , and additional plaintext characters are written into the newly exposed cells.

The openings are arranged so that, across the four orientations (the original position and three rotations), every cell of the square is used exactly once and no two openings overlap.

Once the entire grid has been filled, the characters are read according to a predefined rule to produce the ciphertext.

Cardano grilles are commonly constructed using square grids such as:

4×4 , 5×5 , 6×6 , ...

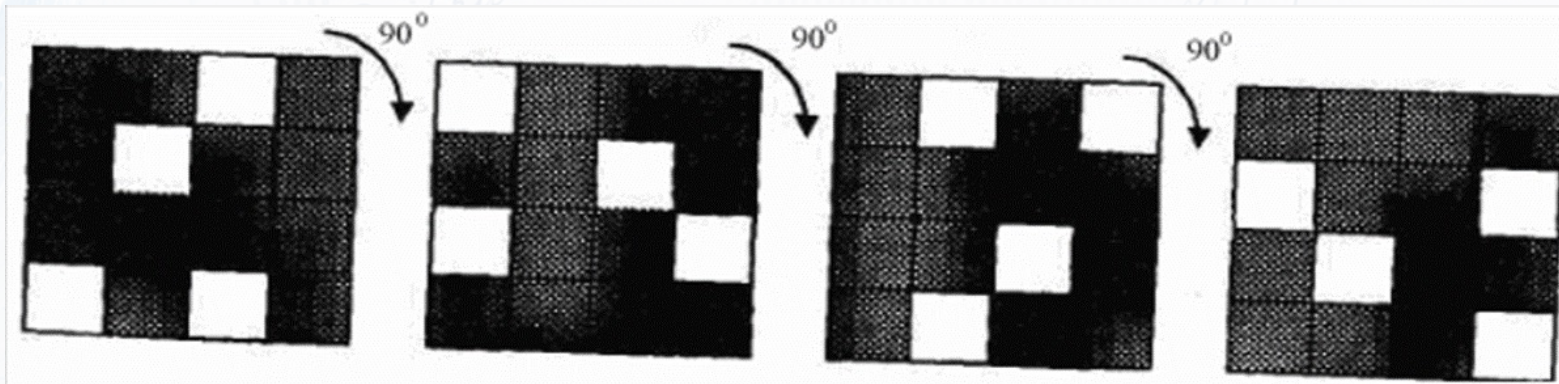


TRANSPPOSITION CIPHER – CARDANO GRILLE

Encryption Using a Cardano Grille

During encryption, the Cardano grille is placed over the writing grid in its initial orientation, and the first portion of the message is written into the cells visible through the openings. The grille is then rotated successively by 90° , 180° , and 270° , and subsequent portions of the message are written into the newly exposed cells.

This process continues until all cells of the grid have been filled. Finally, the characters in the completed grid are read according to a predefined convention to produce the ciphertext.



CRYPTANALYSIS OF THE CARDANO GRILLE

Cryptanalysis of the Cardano Grille

To decrypt a message, the recipient must possess an exact copy of the Cardano grille used during encryption. Consequently, the positions of the openings in the grille constitute the secret key of the cryptosystem.

For a rotating $N \times N$ Cardano grille, the number of possible keys is:

$$|\mathcal{K}| = 4^{N^2/4}$$

where N is even.

For example, for a 6×6 grille:

$$|\mathcal{K}| = 4^{36/4} = 4^9 = 262\,144 = 2^{18}$$

which corresponds to an 18-bit key space.

Although the number of possible keys may be considerable for larger grids, the Cardano Grille is no longer regarded as secure against modern cryptanalytic techniques.

Since it is a purely transposition-based cipher, many statistical and linguistic properties of the plaintext remain preserved in the ciphertext. A cryptanalyst may exploit:

- Common bigrams,
- Common trigrams,
- Frequent word patterns,
- Average word lengths,
- Phonetic and grammatical characteristics of the language,

to gradually recover the underlying permutation and reconstruct the original plaintext.

COLUMNAR TRANSPOSITION CIPHER

Columnar Transposition Cipher

In a Columnar Transposition Cipher, the letters of a keyword are written across the top row of a table. Each letter of the keyword defines a column of the table.

The plaintext is written into the table row by row from left to right. Once the table has been filled, the columns are reordered according to the alphabetical order of the keyword letters.

The ciphertext is then obtained by reading the columns from top to bottom after the column rearrangement has been performed.

To avoid ambiguity during the column ordering process, the keyword is typically chosen so that it contains no repeated letters.

Key:

CALL

Plaintext:

SAIGON – VIETNAM

C	A	L	L
S	A	I	G
O	N		-
	V	I	E
T	N	A	M

L	L	A	C
I	G	A	S
	-	N	O
I	E	V	
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PhD. Tran

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